

Pharmacy Medicine Sales & Billing Management System

Sequential Logic Flow & Standard Diagram Components for Practical Records

Step / Element	Standard Flowchart Symbol	Logical Process / Description
1. START	Oval	System initialization and boot-up.
2. Login	Rectangle	User authentication via credentials.
3. Dashboard	Rectangle	Main panel displaying navigation options.
4. Medicine Details	Parallelogram	Input: Enter or scan medicine details.
5. Customer Details	Parallelogram	Input: Enter customer info (Name, Phone).
6. Check Expiry Date	Rectangle	Retrieve and evaluate product expiry.
7. Is Medicine Expired?	Diamond (Decision)	YES: Go to Reject Sale & Return. NO: Continue Sale.
8. Check Stock	Rectangle	Verify requested batch quantity in system.
9. Is Stock Available?	Diamond (Decision)	YES: Continue Billing. NO: Show Stock Error & Return.
10. Enter Quantity	Parallelogram	Input: Input total items purchased.
11. Calculate Bill	Rectangle	Formula: Quantity × Unit Price.
12. Apply Discount & Tax	Rectangle	Deduct percentage discounts; append GST.
13. Grand Total	Rectangle	Compute final payable amount.
14. Enter Amount Paid	Parallelogram	Input: Cache cash/digital payment received.
15. Calculate Balance	Rectangle	Formula: Amount Paid - Grand Total.
16. Generate Invoice	Parallelogram	Output: Print or display transaction receipt.
17. Update Stock	Rectangle	Inventory: Deduct sold units from DB.
18. Save History	Rectangle	Log full transaction data to database.
19. Generate Reports	Parallelogram	Output: Compile analytical sales reports.
20. END	Oval	Termination of the current transaction.

Pharmacy Sales & Billing Management System (ERD)

1. Visual ERD Component Layout (For Paper Drawing)

Entities (Rectangles): MEDICINE, CUSTOMER, SALE, SALE_DETAILS, EXPIRY_MANAGEMENT

Attributes (Ovals): Connected to respective entities. Underline the Primary Keys (PK).

Relationships (Diamonds) & Cardinality:

- [CUSTOMER] -- (1) -- <Places> -- (M) -- [SALE]
- [SALE] -- (1) -- <Has> -- (M) -- [SALE_DETAILS]
- [MEDICINE] -- (1) -- <Includes> -- (M) -- [SALE_DETAILS]
- [MEDICINE] -- (1) -- <Monitors> -- (1) -- [EXPIRY_MANAGEMENT]

2. Data Dictionary & Entity Attributes

Entity	Attributes (*PK = Primary Key, #FK = Foreign Key)	Description
MEDICINE	<u>Medicine_ID</u> *, Medicine_Name, Company_Name, Batch_Number, Stock, Selling_Price, Purchase_Price, Manufacturing_Date, Expiry_Date	Stores drug stock, pricing, and batch details.
CUSTOMER	<u>Customer_ID</u> *, Customer_Name, Mobile_Number, Address	Stores client profiles and contact records.
SALE	<u>Sale_ID</u> *, Invoice_Number, Customer_ID#, Sale_Date, Subtotal, Discount, GST/Tax, Grand_Total, Amount_Paid, Balance	Tracks structural billing & main finances.
SALE_DETAILS	<u>Sale_Detail_ID</u> *, Sale_ID#, Medicine_ID#, Quantity, Selling_Price, Item_Total	Line-item breakdown for each sale invoice.
EXPIRY_MGMT	<u>Expiry_ID</u> *, Medicine_ID#, Expiry_Date, Status, Alert_Date	Handles proactive shelf-life drug alerts.

3. Cardinality & Relationship Rules

- Customer → Sale (1:M):** One customer can have multiple sales invoices, but each sale belongs to one customer.
- Sale → Sale_Details (1:M):** One sale transaction contains multiple medicine line items detailed individually.
- Medicine → Sale_Details (1:M):** A medicine can be sold across multiple separate sales transactions over time.
- Medicine → Expiry_Management (1:1):** Each unique medicine record links strictly to one expiry status tracking entry.

Pharmacy Medicine Sales and Billing Management System Algorithm

M1: Authentication & Menu

1. **START**
2. **INPUT:** Read Username, Password.
3. **DECISION:** If credentials valid, show Main Menu. Else, re-prompt (max 3 times) then fail.
4. **INPUT:** Read MenuChoice (1: Medicines, 2: Customers, 3: Quality, 4: Billing, 5: Reports, 6: Exit).

M2: Medicine Inventory Management

5. **INPUT:** Select Operation (Add / Update / Delete / Search).
6. **PROCESS (Add):** INPUT details → Append to Medicine_Database.
7. **PROCESS (Update):** Find Medicine_ID → Modify stock/price fields.
8. **PROCESS (Delete):** Find Medicine_ID → Remove entry from records.
9. **PROCESS (Search):** Query string match → OUTPUT item details.

M3: Customer Management

10. **INPUT:** Select Operation (Add / Update / Delete / Search).
11. **PROCESS (Add):** INPUT demographics → Append to Customer_Database.
12. **PROCESS (Update/Delete):** Find Customer_ID → Modify/Erase profile.
13. **PROCESS (Search):** Query Phone/ID → OUTPUT customer profile.

M4: Quality Control & Expiry Checks

14. **PROCESS:** Set Expiry_Threshold = Current_Date + 90 Days.
15. **DECISION:** Parse Medicine_Database. If Expiry_Date ≤ Current_Date, status = 'EXPIRED'. Else if Expiry_Date ≤ Expiry_Threshold, status = 'NEAR EXPIRY'.
16. **DECISION:** If Available_Stock == 0, OUTPUT 'Out of Stock' alert.

M5: Billing & Order Processing

17. **PROCESS:** Initialize Subtotal=0, Tax=0, Discount=0, Grand_Total=0.
18. **INPUT:** Read Customer_ID. Link profile or fallback to 'Walk-in'.
19. **INPUT:** Scan Medicine_ID.
20. **DECISION:** If item status == 'EXPIRED', OUTPUT 'SALE BLOCKED' → Skip item.
21. **INPUT:** Read Sale_Quantity.
22. **DECISION:** If Sale_Quantity > Available_Stock, OUTPUT error → Re-prompt.
23. **PROCESS:** Calculate Item_Total = Sale_Quantity × Selling_Price.
24. **PROCESS:** Add to Sales_Cart; update Subtotal = Subtotal + Item_Total.
25. **INPUT:** Query: 'Add more items?'. If YES, loop back to Step 19.
26. **INPUT:** Read Discount_%, Tax_% rates.
27. **PROCESS:** Compute financial breakdown equations:
 - $\text{Discount} = \text{Subtotal} \times (\text{Discount_}\% / 100)$
 - $\text{Taxable_Amt} = \text{Subtotal} - \text{Discount}$
 - $\text{Tax} = \text{Taxable_Amt} \times (\text{Tax_}\% / 100)$
 - $\text{Grand_Total} = \text{Taxable_Amt} + \text{Tax}$
28. **INPUT:** Read Amount_Paid. If Amount_Paid < Grand_Total, loop error.
29. **PROCESS:** Balance_Change = Amount_Paid - Grand_Total.
30. **PROCESS:** For each item in cart, deduct stock: Available_Stock = Available_Stock - Sale_Quantity.
31. **PROCESS:** Commit payload to Customer_Purchase_History and log revenue transactions into Daily_Sales_Journal.
32. **OUTPUT:** Print generated itemized system invoice.

M6: Business Reports & Shutdown

33. **INPUT:** Select ReportType (Daily / Monthly).
34. **PROCESS:** Compile summary metrics from Daily_Sales_Journal.
35. **OUTPUT:** Display gross total sales revenue figures.
36. **PROCESS:** Commit state updates safely to non-volatile disk arrays.
37. **END**